# Who Bears the Burden of Bank Capital Regulation?

## Buffers, Risk Weights, and the Supply of Credit

Juan Zurita

School of Economics, University of Edinburgh

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# Motivation

**Why capital requirements exist:** equity absorbs unexpected losses, so banks stay solvent in downturns.

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stability benefits of a higher requirement vs. the output cost of getting there

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- Every revision redraws the line on the same dimension: *asset size*.
- Both margins presume regulators know *where a tightening binds*.

# Research Question

*Who bears the burden of a capital requirement tightening?*

- Which **sectors** absorb the contraction in credit?
- Which **banks** transmit it?
- What is the **aggregate cost**—and does design matter for it?

# What I Do

**Empirics: measure who bears a tightening**

- US Call Reports, 4,800 banks, 1994–2019; identified capital-requirement shocks; within-bank local projections
- Two margins: **across *sectors*** (C&I vs. real estate), and **across *banks*** by distance to their own requirement

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**Quantitative model: the same two margins, solved**

- A continuum of banks, each choosing its own distance to its requirement; the capital constraint binds occasionally
- Estimated by SMM on the *sectoral* responses only — the cross-section is left free

**Technique:** policy rules and shadow prices parameterised by deep neural networks

Why global Solution method

# What I Find

**Empirics: two facts**

**1. Across sectors:** business (C&I) lending contracts **1.3**× as much as real estate

**2. Across banks:** banks with the least room above *their own* requirement cut **1.3**× as much as those with the most

**Quantitative model: fitted to fact 1, it reproduces fact 2**

- **Across banks:** **1.2**× in the model vs. **1.3**× in the data — a moment the estimation never used

The burden falls on banks with the least room above their own requirement.

*Level* of requirements ⇒ aggregate cost. *Design* ⇒ **incidence**.

# What's New: the Buffer Distribution

**What:** each bank's ***distance to its own requirement*** — the buffer $D_{i,t} \equiv \mathrm{CR}_{i,t} - \kappa_{i,t}$, in percentage points

- What it costs a bank to raise wholesale funding ($\gamma_i$) makes that distance a *choice*: **costlier funding, thicker *target* buffer**
- **This paper solves at every point of the buffer distribution — not one intermediary, not two types**

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**Context:** the closest quantitative models do not carry that distribution

- Elenev, Landvoigt, and Van Nieuwerburgh (2021): one intermediary ⇒ incidence is sector-level by construction
- Corbae and Erasmo (2021): two types, buffers matched at two group means

Proposition 1 Related lit.

# Roadmap: the Paper and This Talk

**The paper has seven sections. This talk covers five.**

1. Introduction ✓
2. Empirical Evidence ✓
3. A Stylized Model
4. Quantitative Model ✓
5. Quantitative Results ✓
6. Discussion
7. Conclusion ✓

# Funding Capacity and the Buffer

- **Buffer** $D_{i,t} \equiv \mathrm{CR}_{i,t} - \kappa_{i,t}$ (distance to its own requirement, pp)

  $\mathrm{CR}_{i,t}$: capital ratio (regulatory capital / RWA). $\kappa_{i,t}$: bank $i$'s requirement ($\kappa_t$ plus its statutory tier).

- **Funding Capacity** $\equiv$ (core retail + NIB deposits) / total liabilities

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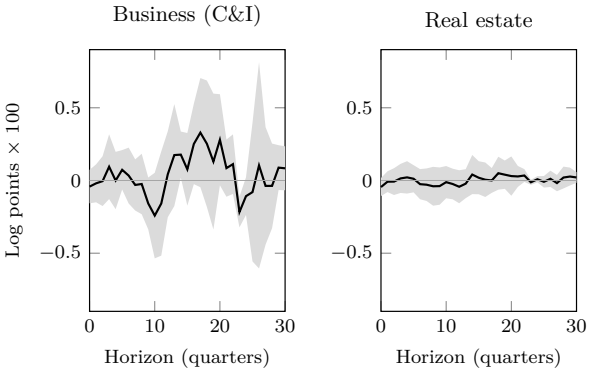
**Across banks of *similar size*, those that rely on wholesale funding hold *thicker* buffers; those with stable deposit franchises hold *thinner* ones.**

Regulation | Distribution | Regression | Leverage vs. RWA | Buffer by size | Proposition 1

# Two Empirical Facts

# Fact 1: Sectoral Asymmetry in Credit Responses

**Identification:** Narratively identified capital-requirement shocks (Drechsel and Miura, 2025); within-bank local projections



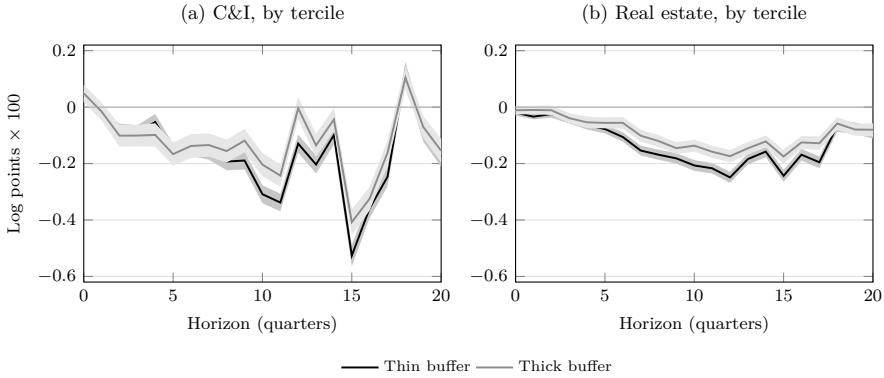

**Scale Deleveraging Multiplier**

$$\frac{\partial \text{RWA}}{\partial A^B} = \omega_B > \omega_M = \frac{\partial \text{RWA}}{\partial A^M}$$

Each dollar of C&I carries twice the regulatory cost per dollar lent — paid *whether or not* the constraint binds.

All four | Identification | Interpretation

# Fact 2: Thin-Buffer Banks Cut More

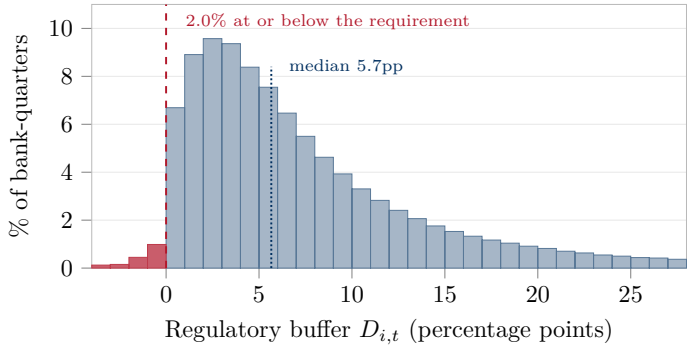



Banks sorted by lagged buffer $D_{i,t-1}$.

Thin-buffer banks contract 1.32× more in C&I — and 1.41× in real estate.

Details Precision

# The Buffer Distribution in the Data




$D_{i,t} \equiv \mathrm{CR}_{i,t} - \kappa_{i,t}$, with $\kappa_{i,t} = 10\%$ for every bank. Call Reports, 1993Q2–2019Q4: 940,519 bank-quarters, 16,162 banks, 1pp bins.

# What the Buffer Gradient Identifies

**A capital constraint alone cannot produce it.**

- *Every* tercile contracts — even *thick*-buffer banks, furthest from their requirement, cut significantly ($-0.13\%$)
- Yet fewer than 3% of bank-quarters are ever *at* their requirement — so a binding constraint cannot be what transmits the tightening

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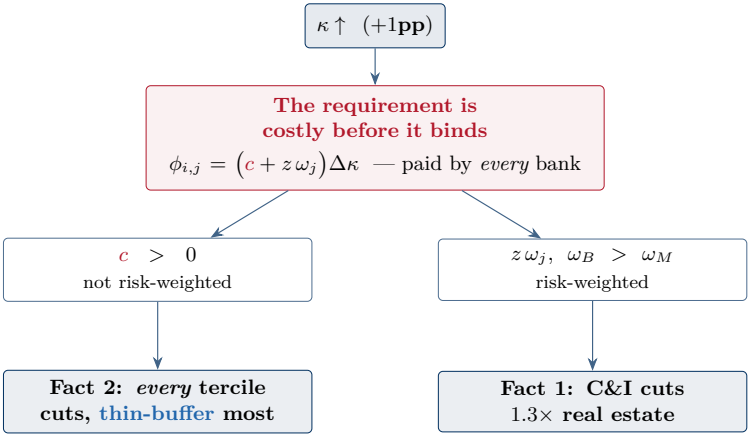
**So the requirement is costly *before* it binds:**

$$\phi_{i,j}(\kappa) \;=\; \big(c + z\,\omega_j\big)\,\max(\kappa - \bar{\kappa}_i,\,0)$$

- The gradient identifies that this cost *exists*; the *sectoral* split identifies its composition. $c > 0$ denotes the cost paid per dollar lent whatever the risk weight, while $z\,\omega_j$ is paid only through it

Facts recap  What $c$ and $z$ are

# The Mechanism




One term, both facts: $c$ sets the level, $z\,\omega_j$ the sectoral split.

# Model

# Model: Environment

**Dynamic, partial equilibrium, heterogeneous-agent.**

- Continuum of infinitely lived, risk-neutral banks; equity state $e_{i,t}$
- Two sectors—real estate ($j = M$) and C&I ($j = B$)—with inverse loan demand $r_{j,t} = \theta_j - \eta_j L_t^j$

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- Retail deposits are limited — the marginal dollar comes from wholesale funding $d_{i,t}^W$, at quadratic cost $\frac{\gamma_i}{2}(d_{i,t}^W)^2$
- $\gamma_{\text{small}} \gg \gamma_{\text{large}} \to 0$: small banks face steep wholesale costs — this is where buffer heterogeneity comes from

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- $\gamma_{\text{small}} \gg \gamma_{\text{large}} \to 0$: small banks face steep wholesale costs — this is where buffer heterogeneity comes from
- One aggregate disturbance: $\kappa_t$ follows an AR(1); bank $i$ faces $\kappa_{i,t} = \kappa_t +$ its legal tier add-on
- Two occasionally binding constraints: regulatory capital, and dividend non-negativity

Market clearing Stylized model

# The Dynamic Bank Problem (I): The Problem

**Banks choose lending and dividends to maximise the discounted dividend stream:**

$$V(e_{i,t}, A^M_{i,t-1}, A^B_{i,t-1}; \Omega_t) = \max_{\{A^M_{i,t},\, A^B_{i,t},\, c_{i,t}\}} \Big\{ c_{i,t} + \beta\, \mathbb{E}_t\, V\big(e_{i,t+1}, A^M_{i,t}, A^B_{i,t}; \Omega_{t+1}\big) \Big\}$$

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**Subject to the balance-sheet identity** — wholesale funding is *residual*, not chosen:

$$d^W_{i,t} = A^M_{i,t} + A^B_{i,t} - e_{i,t} - d_{i,C}$$

and to the law of motion for equity, which carries returns, costs and the levy into $e_{i,t+1}$.

Equity law of motion

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**and to two occasionally binding constraints:**

$$\underbrace{\omega_M A^M_{i,t} + \omega_B A^B_{i,t} \le \frac{e_{i,t}}{\kappa_{i,t}}}_{\text{regulatory capital}} \iff \underbrace{D_{i,t} \ge 0}_{\text{the buffer of Fact 2}} \quad , \quad \underbrace{c_{i,t} \ge 0}_{\text{no equity issuance}}$$

Euler equation · Solution method · Calibration

# What the Model Adds

**The local projections give responses. The model gives three things they cannot:**

- **Incidence** — the burden *per dollar lent*, by buffer position
- **A price** — the welfare cost of a tightening, as a share of credit-market surplus
- **Counterfactuals** — a uniform requirement, or flat risk weights: regimes the data never ran

# SMM Estimation Strategy

**Estimate:** $\theta = (\rho_\kappa, \sigma_\kappa)$ — persistence and volatility of the AR(1) capital-requirement process

**Targets:** 16 within-bank LP coefficients
C&I and RE responses at $\mathcal{H} = \{1, 2, 4, 6, 8, 12, 16, 20\}$

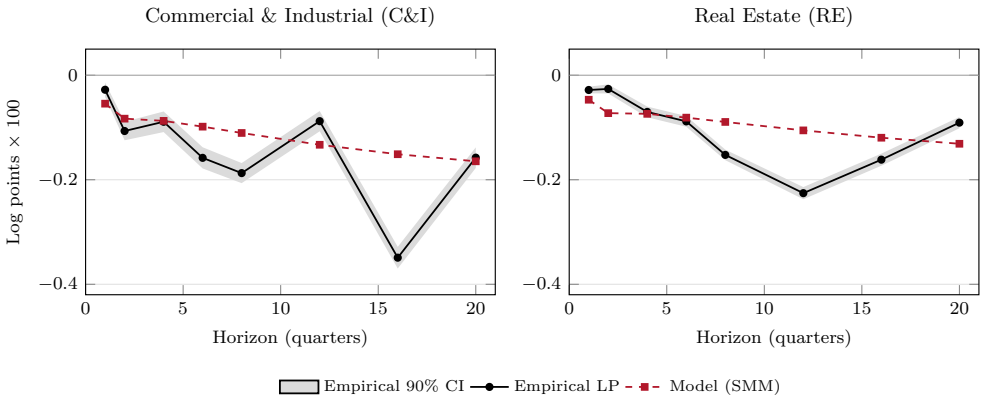
**Identification, in words:**

- $\rho_\kappa \leftarrow$ how slowly the IRF decays at medium horizons
- $\sigma_\kappa \leftarrow$ how large the impact response is
- Two parameters, sixteen moments

Weighting and algorithm Calibration

# Estimation: Model vs. Data

**The 16 targeted moments: sectoral credit responses, model against data.**




$\hat{\rho}_\kappa = 0.9987$: a tightening is not reversed — 97% of it is still in place at horizon 20.

Moment fit | Robustness

# A Prediction, Not a Fit

**Targeted** — the 16 moments in the previous figure:

sectoral credit responses, C&I and real estate, at eight horizons

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**Not targeted** — which banks cut most:

| Thin-to-thick response ratio | data (90% CI) | model |
|---|---|---|
| C&I | 1.32 [1.01, 1.62] | **1.23** |
| Real estate | 1.41 [1.14, 1.67] | **1.18** |

Both model values lie *inside* the data's 90% band. Model column: the $\kappa$-augmented *dynamic* solve; the static closed form gives 1.32 and 1.08.

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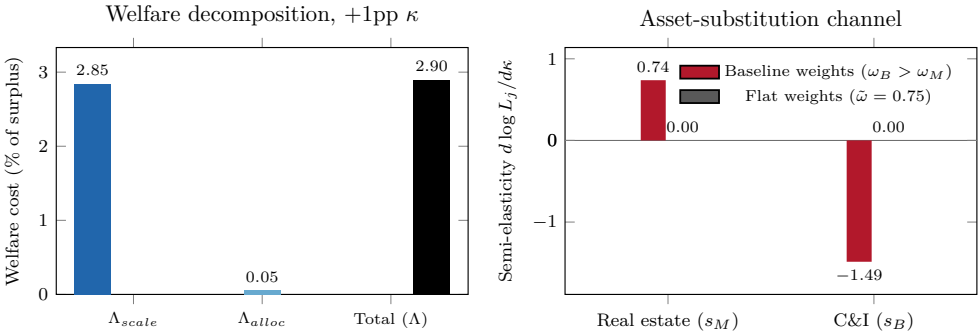
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- The model was fitted to *how much* credit falls — never to *which banks* cut most.

Precision

# Gross Cost and Decomposition



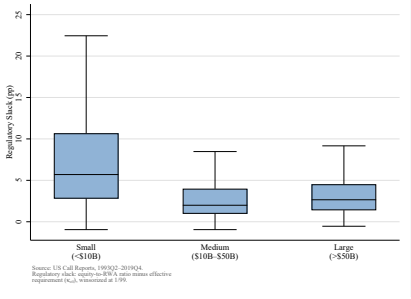

Left: CEV decomposition. Right: constrained-regime semi-elasticities.

Counterfactuals: neither a uniform $\tilde{\kappa}$ nor flattened risk weights changes the aggregate burden — but both change **who bears it**.

Counterfactuals | Details | CEV framework | Magnitudes in context

# Who Bears It? Incidence Across Banks

**The regulation sorts by size. The mechanism sorts by buffer.**

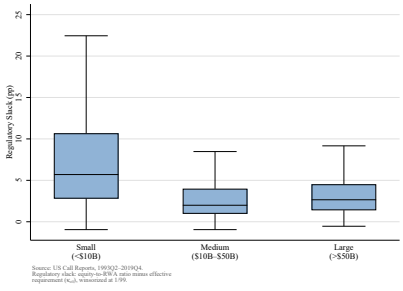



Buffers vary as much *within* each statutory tier as across them.

How it's measured | Tier mix | By tier | Over time | Buffer by size

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**In the model, a +1pp tightening:**

- **Thin-buffer banks:** 32% of lending, 34% of the burden
- **Thick-buffer banks:** 35% of lending, 33% of the burden

**Per dollar lent, thin-buffer banks bear 13% more than thick-buffer banks.**

How it's measured | Tier mix | By tier | Over time | Buffer by size

# Conclusion

**Who bears the burden of bank capital regulation?**

- **Which sectors:** risk weights concentrate the contraction in C&I — 1.31× the real-estate response
- **Which banks:** the buffer, not size — thin-buffer banks cut 1.32× as much in the data
- **And the law does not see it:** the statute sorts by size; the burden sorts by distance to the requirement

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*Design* ⇒ **who pays**. *Level* ⇒ the aggregate cost.

1.4–2.4% of credit-market surplus per +1pp, gross of the stability benefits of capital.

# Thank you

Who Bears the Burden of Bank Capital Regulation?

Juan Zurita | University of Edinburgh

juan.zurita@ed.ac.uk

# Interpreting the Sectoral Asymmetry

**Why C&I contracts more than RE:**

- **Asset Substitution Margin:** when the constraint binds, banks shed high-$\omega$ assets (C&I, $\omega_B = 1$) to free the most required capital per dollar cut

- **Scale Deleveraging Multiplier:** $\partial \text{RWA}/\partial A^B = \omega_B > \omega_M = \partial \text{RWA}/\partial A^M$
Each dollar of C&I cut yields twice the regulatory relief of a mortgage cut

- **And away from the frontier:** $\partial \phi_{i,j}/\partial \kappa = c + z\,\omega_j > 0$, so the same ordering holds for *every* bank, constrained or not — which is why the asymmetry is not confined to the <3% at the frontier

- **Consumer credit unresponsive:** high interest margins provide sufficient cash-flow buffers to absorb the regulatory cost without volume rationing

**Result:** the sectoral composition of the contraction is determined by the risk-weight schedule, not by loan demand.

Back

# SMM Moment Fit

| Sector | Horizon $h$ | $\hat{b}_h$ (data) | s.e. | $b_h$ (model) | $t$ |
|---|---|---|---|---|---|
| C&I | 1 | -0.028 | 0.008 | -0.054 | +3.20 |
| | 2 | -0.107 | 0.011 | -0.083 | -2.20 |
| | 4 | -0.089 | 0.012 | -0.087 | -0.14 |
| | 6 | -0.158 | 0.012 | -0.098 | -4.85 |
| | 8 | -0.187 | 0.012 | -0.110 | -6.57 |
| | 12 | -0.088 | 0.012 | -0.133 | +3.84 |
| | 16 | -0.349 | 0.013 | -0.151 | -15.73 |
| | 20 | -0.158 | 0.012 | -0.165 | +0.58 |
| RE | 1 | -0.028 | 0.004 | -0.047 | +4.87 |
| | 2 | -0.026 | 0.005 | -0.073 | +8.80 |
| | 4 | -0.070 | 0.006 | -0.074 | +0.59 |
| | 6 | -0.088 | 0.006 | -0.081 | -1.15 |
| | 8 | -0.152 | 0.006 | -0.089 | -10.29 |
| | 12 | -0.226 | 0.007 | -0.106 | -17.02 |
| | 16 | -0.162 | 0.007 | -0.120 | -5.97 |
| | 20 | -0.091 | 0.007 | -0.131 | +6.16 |
| Hansen $J = 1822.15$, dof $= 14$, $\chi^2_{0.05} = 23.68$, $p = 0.000$ | | | | | |

Signs correct for all 16 moments; sectoral ordering reproduced at every horizon. $J = 1822$ (14 d.f.) reflects the high precision of the bank panel—the economically relevant misfit is in the trough depths.

Back

# Identification: Capital-Requirement Shocks

**Shock series:** Drechsel and Miura (2025)

- Narrative + sign-restriction approach
- Captures unanticipated changes in the capital-requirement stance
- Available 2002Q1–2023Q2; estimation sample capped at 2019Q4

**Local Projection specification:**

$$x_{t+h} - x_{t-1} = \alpha_h + \beta_h \, \text{shock}_t^{\text{Cap}} + \sum_{p=1}^{6} \Psi_{h,p} \, x_{t-p} + \varepsilon_{t+h}$$

- **Aggregate:** OLS, Newey–West (lag $h + 6$)
- **Bank-level:** bank FE, clustered by cert
- Consumer credit placebo: no response $\Rightarrow$ supply, not demand

Back

# The Buffer Gradient: Details

- Thin-buffer banks cut C&I 1.32× more than thick-buffer banks (90% CI [1.01, 1.62]); 1.41× in real estate, 1.37× pooled [1.13, 1.61] — from a joint cluster-robust covariance across the separate tercile regressions

- The risk-weight mechanism implies the *same* gradient in both sectors, and the restriction is not rejected: difference +0.090 (s.e. 0.186), $t = 0.48$

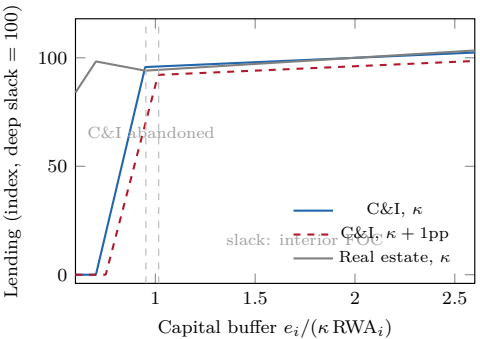
- Freezing tercile membership at each bank's pre-sample average slack **reverses** the gradient, to 0.67 in C&I — it is the buffer a bank holds when the shock arrives that matters, not a fixed bank type

Horizon profile, cohort power, and the continuous-interaction specification: see backup.

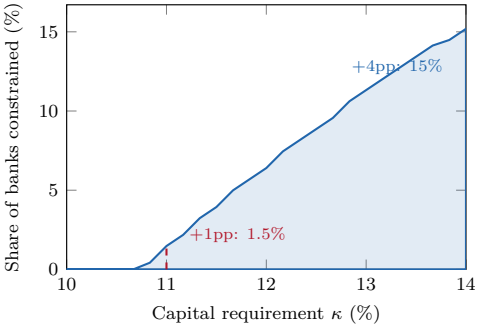
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# Why a Global Non-Linear Method

(a) Policy functions are kinked at the frontier




(b) Transmission depends on the mass at the kink




Both panels are the *exact* static solution — the problem is quadratic with a linear constraint, so it solves in closed form in each KKT regime.

What the kink implies Back

# Why a Global Non-Linear Method

- The capital constraint binds occasionally, so the policy function is **kinked**: left of the frontier lending is pinned by equity, right of it the interior FOC holds and lending is flat in equity
- A +1pp tightening moves the kink *right* — banks that were unconstrained become constrained
- So the aggregate response is not a fixed elasticity: it depends on the **mass sitting near the kink**. +1pp constrains 1.5% of banks, +4pp constrains 15%

The figure Back

# Backup: Neural-Network Solution

**Challenge:** occasionally binding constraints ⇒ kinked policy functions.

**Approach:** parameterise policy with a multi-layer perceptron (MLP):

$$\begin{bmatrix} \hat{A}^M_{i,t} \\ \hat{A}^B_{i,t} \\ \hat{c}_{i,t} \\ \hat{\lambda}_{i,t} \\ \hat{\mu}_{i,t} \end{bmatrix} = \mathcal{A}(S_{i,t}; \Theta) \qquad S_{i,t} \equiv (e_{i,t}, A^M_{i,t-1}, A^B_{i,t-1}, \Omega_t)$$

$\Omega_t \equiv (\kappa_t,\, L^M_t,\, L^B_t)$: the current requirement and the two sectoral credit aggregates.

Network Method Non-convexity Accuracy What it pins down Back

# Backup: Solution Method — Implementation

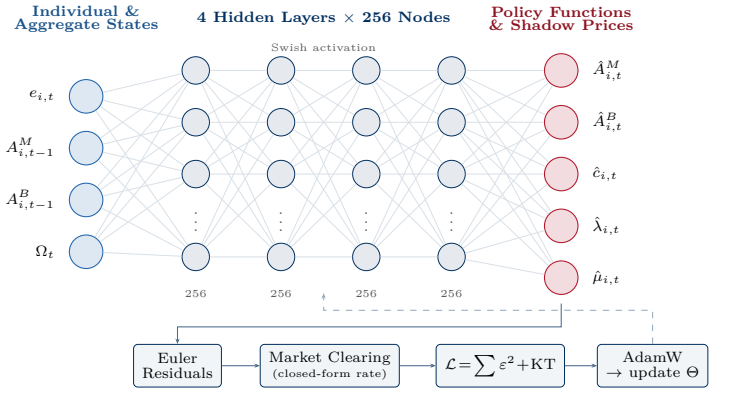
**How the policy is trained.**

- **Structural loss:** squared Euler residuals + soft KT penalty
- **Market clearing:** imposed analytically via closed-form equilibrium rate
- **Architecture:** 4 hidden layers $\times$ 256 nodes, Swish activation
- **Optimizer:** AdamW with cosine annealing

The Karush–Kuhn–Tucker conditions enter as a soft penalty rather than being imposed by construction; the multiplier uses the cubed García–Zangwill substitution, which makes complementarity hold identically and keeps the map twice differentiable at the kink.

Back to network Back to solution

# Neural-Network Architecture




*Training: ∼6h on NVIDIA A100 · Cosine annealing schedule · Euler errors $< 10^{-3}$ across state space*

Implementation Back

# Gross Cost: Numbers

| Experiment | Quantity | Value |
|---|---|---|
| *CEV of +1pp $\kappa$ tightening (eqs. (??)–(??))* | | |
| Total welfare cost | $\Lambda$ | 1.421% |
| scale channel | $\Lambda_{scale}$ | 1.415% |
| allocation channel | $\Lambda_{alloc}$ | 0.004% |
| additivity check | $\Lambda_{scale} + \Lambda_{alloc}$ | 1.419% |
| Credit contraction (total / RE / C&I) | $\Delta L$ | -3.09% / -3.02% / -3.50% |
| *CEV from the franchise value $V$ (discounted dividends)* | | |
| Total welfare cost | $\Lambda_V$ | 2.404% |
| scale channel | $\Lambda_{V,scale}$ | 2.392% |
| allocation channel | $\Lambda_{V,alloc}$ | 0.026% |
| additivity check | sum | 2.419% |
| *CF1: uniform vs. size-differentiated $\kappa$ (eq. (??))* | | |
| Differentiated rule | $\Lambda$ | 1.236% |
| Uniform $\tilde{\kappa} = 0.1246$ | $\Lambda$ | 1.239% |
| *CF2: flat risk weights $\tilde{\omega} = 0.75$ (eq. (??))* | | |
| Baseline sectoral asymmetry | $s_B - s_M$ | -2.224 |
| Flat-weight sectoral asymmetry | $s_B - s_M$ | -0.000 |

Credit-market surplus: $\Lambda = 1.4\%$. Franchise value: $\Lambda_V = 2.4\%$. Scale channel accounts for >99% of the cost. Magnitudes are *directional*: robust in sign and decomposition, sensitive to the solved policy. The cross-sectional gradient is not.

Back

# Policy Counterfactuals

**CF1: Uniform vs. size-differentiated $\kappa$**

- Replace the step function with a uniform $\tilde{\kappa} = 0.1246$ raising the same aggregate capital
- Gross cost: 2.680% vs. 2.682% $\Rightarrow$ size-tailoring redistributes but barely reduces the aggregate burden

**CF2: Risk-weight flattening**

- Set $\tilde{\omega}_B = \tilde{\omega}_M = 0.75$ (average), hold $\kappa$ fixed
- Sectoral asymmetry $s_B - s_M$ collapses from $-2.22$ to 0.00
- Aggregate credit contraction essentially unchanged

**Level** of requirements $\Rightarrow$ aggregate cost.
**Design** (risk weights, size tiers) $\Rightarrow$ incidence.

Details Back

# Backup: Counterfactual Definitions

**CF1 — uniform vs. size-differentiated requirements.** Replace the statutory step function $\kappa_i(\text{Scale}_i)$ with the single rate $\tilde{\kappa}$ that raises the *same aggregate capital* in the stationary distribution $\mu^*$:

$$\tilde{\kappa} \;\text{ s.t. }\; \int \tilde{\kappa}\, e_i \,\mathrm{d}\mu^* \;=\; \int \kappa_i(\text{Scale}_i)\, e_i \,\mathrm{d}\mu^*.$$

So the experiment holds the *level* of regulation fixed and varies only its *design*: any difference in $\Lambda$ is pure incidence.

**CF2 — flat risk weights.** Set $\omega_M = \omega_B = \tilde{\omega} = 0.75$, removing the risk-weight wedge. The test is whether the two sectors then deleverage proportionally:

$$\frac{\partial \tilde{L}^B_{t+h}}{\partial \varepsilon^{\kappa}_t} \;\approx\; \frac{\partial \tilde{L}^M_{t+h}}{\partial \varepsilon^{\kappa}_t} \quad \forall\, h.$$

| | $\Lambda$ | sectoral asymmetry $s_B - s_M$ |
|---|---|---|
| Differentiated (baseline) | 1.236% | $-2.224$ |
| Uniform $\tilde{\kappa} = 0.1246$ | 1.239% | — |
| Flat weights $\tilde{\omega} = 0.75$ | — | $-0.000$ |

Design redistributes the burden; it barely moves the aggregate. Flattening the weights eliminates the sectoral asymmetry entirely.

Back

# Backup: Related Literature

## Quasi-experimental credit supply

- Aiyar, Calomiris, and Wieladek (2016), Gropp, Guettler, and Saadi (2020), De Jonghe, Dewachter, and Ongena (2020), Jimenez et al. (2017), Conti, Nobili, and Signoretti (2023)
- Here: the same identified shock is traced across sectors *and* across the buffer distribution, then mapped into a structural cost.

## Deposit franchise and regulatory incidence

- Drechsler, Savov, and Schnabl (2017)
- The deposit franchise shapes not only monetary transmission but the incidence of *capital* regulation: what disciplines the buffer is the *marginal* funding source.

## Quantitative macro-banking

- Corbae and Erasmo (2021), Begenau (2020), Elenev, Landvoigt, and Van Nieuwerburgh (2021), Mendicino et al. (2020)

Back

# Backup: Reading $c$ and $z$

$$\phi_{i,j}(\kappa) \;=\; \big(c + z\,\omega_j\big)\,\max(\kappa - \bar{\kappa}_i,\,0)$$

- $z = 0.021$ is **risk-weighted**: a Modigliani–Miller premium, $\approx 10\%$ annualised equity–debt spread
- $c = 0.148$ is **not** risk-weighted and seven times larger — what a **leverage requirement** imposes. U.S. banks face one; the model's constraint is risk-based only
- Both are set on the two sectoral aggregates; *both* buffer gradients are untargeted

The leverage-requirement reading is a conjecture about the *source* of the non-risk-weighted term. The data identify its *magnitude* either way.

Back

# Backup: SMM Weighting and Algorithm

**Weighting matrix**

$\mathbf{W} = \hat{\Sigma}^{-1}$, block-diagonal, estimated by cluster bootstrap of the bank-level local projections (100 replications)

**Algorithm — for each candidate $\theta = (\rho_\kappa, \sigma_\kappa)$**

1. Re-solve the model globally with the neural network
2. Simulate a panel impulse path under the candidate process
3. Compute model-implied LP coefficients on the simulated panel
4. Evaluate $(\hat{m} - m(\theta))' \mathbf{W} (\hat{m} - m(\theta))$

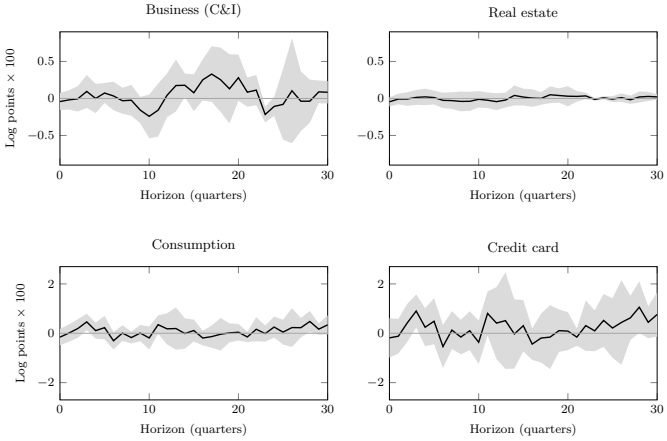
Each evaluation requires a full re-solve, which is what makes the global solution method the binding cost of estimation.

Hansen $J$ rejects. With ≈4,800 banks the moments are basis-point precise, so the test weights economically trivial misses heavily: sign, sectoral ordering and hump shape all match; trough depth does not.

Validation is on the *ratio*, not the level: $\sigma_\kappa$ is a free scale, so the depth of any model response is fitted by construction. This is why the untargeted test is stated as a thin-to-thick ratio.

Back

# Backup: All Four Loan Categories




C&I contracts most, real estate less, and consumption and credit-card lending do not respond at any horizon. The two unresponsive categories are the demand placebo: they are fully risk-weighted but funded against high interest margins, so a capital tightening need not be absorbed through volume.

Back

# Backup: Summary of Empirical Facts

| Fact | Finding |
|---|---|
| 1. Sectoral asymmetry | C&I contracts ∼a third more than RE within banks (risk-weight wedge) |
| 2. Buffer gradient | Thin-buffer banks cut 1.32× more; with terciles frozen at pre-sample slack the gradient *reverses* (0.67) |
| 3. Placebo | Consumption and credit-card lending unresponsive at all horizons |
| 4. Robustness | Results stable to 2023Q2 extension, pandemic controls |

Back

# The Dynamic Bank Problem (II): The Euler Equation

**Optimal lending in sector $j$ satisfies:**

$$\beta\,\mathbb{E}_t\left[(1+\mu_{i,t+1})\left(r_{j,t}-\varepsilon^j_{i,t+1}-\psi_j A^j_{i,t}\underbrace{-r_{W,t}-\gamma_i d^W_{i,t}}_{\text{marginal funding}}-\phi_{i,j}(\kappa_{i,t})\right)\right]=\lambda_{i,t}\,\omega_j$$

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- $\lambda_{i,t}$: shadow cost of the capital constraint — zero whenever $D_{i,t}>0$
- $\mu_{i,t}$: multiplier on $c_{i,t}\geq 0$ — internal funds are worth more than one when the bank would cut dividends below zero and cannot

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- $\mu_{i,t}$: multiplier on $c_{i,t}\geq 0$ — internal funds are worth more than one when the bank would cut dividends below zero and cannot
- $\phi_{i,j}$: per-unit cost of regulatory capital, paid *whether or not* the constraint binds (bank-specific via $\bar{\kappa}_i$)
- $\omega_B>\omega_M \Rightarrow$ higher regulatory cost per unit C&I $\Rightarrow$ C&I contracts more

Where these terms come from  Back

# Backup: Where the Euler Terms Come From

**Law of motion for equity.** Every term in the Euler equation is a derivative of this line.

$$e_{i,t+1} = (1-\delta_e)\, e_{i,t} + \sum_{j\in\{M,B\}} \left[ (r_{j,t} - \varepsilon^j_{i,t+1}) A^j_{i,t} - \tfrac{\psi_j}{2}(A^j_{i,t})^2 - \phi_{i,j}(\kappa_{i,t})\, A^j_{i,t} \right]$$
$$- r_{d,t}\, d_{i,C} - r_{W,t}\, d^W_{i,t} - \tfrac{\gamma_i}{2}(d^W_{i,t})^2 - c_{i,t} - \tau_t e_{i,t}$$

**Wholesale funding is residual**, not chosen: $d^W_{i,t} = A^M_{i,t} + A^B_{i,t} - e_{i,t} - d_{i,C}$

$\varepsilon^j_{i,t+1}$: idiosyncratic credit loss, realised after lending, so the net return per dollar is $r_{j,t} - \varepsilon^j_{i,t+1}$

$\psi_j$: quadratic operating cost — marginal cost $\psi_j A^j_{i,t}$ rises with portfolio size, which pins the long-run sectoral mix

$r_{d,t}$, $r_{W,t}$: retail deposit and wholesale funding rates

$\delta_e$: quarterly equity depreciation (operating costs, franchise decay)

$\tau_t$: systemic insurance levy on equity, funding entrant capital injections

Back to Problem (I) Back to Euler (II)

# Backup: Funding Frictions and the Target Buffer

**Proposition 1.** In the stationary equilibrium, a bank with funding friction $\gamma_i$ and core-deposit capacity $\bar{d}_{i,C}$ targets wholesale funding

$$d_i^{W\star} = \frac{\rho + \delta_e + \tau - r_W}{\gamma_i},$$

and holds equity $\bar{e}_i$ satisfying $\bar{e}_i = A_i(\bar{e}_i) - \bar{d}_{i,C} - d_i^{W\star}$. Consequently:

1. the target capital ratio $\bar{e}_i/\mathrm{RWA}_i$ is strictly increasing in $\gamma_i$;
2. there is a threshold $\gamma^*$: banks with $\gamma_i \leq \gamma^*$ target the regulatory frontier *exactly* — zero voluntary buffer — while banks with $\gamma_i > \gamma^*$ target a strictly positive buffer;
3. the elasticity of target equity to the requirement is $\approx 1$ for $\gamma_i \leq \gamma^*$ and $\approx 0$ for $\gamma_i > \gamma^*$.

▸ (i) is the *costlier funding, thicker target buffer* claim; (ii) is why a few banks sit exactly at the frontier and most do not; (iii) says a requirement change moves target equity only for the banks already at the frontier

Back Funding capacity

# Backup: Slack on Funding Capacity

| **Dep. var:** Slack (pp) | (1) Pooled | (2) Between | (3) Within |
|---|---|---|---|
| Funding Capacity | $-19.44^{***}$ | $-25.97^{***}$ | $-2.27^{**}$ |
| | (1.73) | (1.95) | (1.05) |
| Log Assets | $-2.80^{***}$ | $-2.70^{***}$ | $-6.27^{***}$ |
| | (0.09) | (0.10) | (0.16) |
| Quarter FE / Bank FE | Yes / No | — | Yes / Yes |
| $N$ | 935,234 | 16,117 | 935,056 |

The between estimate ($-25.97$) is the cross-sectional regularity; the within estimate ($-2.27$) is small because the funding mix barely moves at quarterly frequency. Size enters separately and negatively throughout, so this is not size in disguise.

Back

# Stylized Model: Environment

**Static, partial equilibrium.** Continuum of risk-neutral banks $i \in [0, 1]$.

- Two asset classes: mortgages ($A_i^M$, $\omega_M = 0.50$) and C&I ($A_i^B$, $\omega_B = 1.00$)
- Liabilities: core deposits $d_{i,C}$ (cheap, limited) + wholesale $d_{i,W}$ (quadratic friction $\frac{\gamma_i}{2} d_{i,W}^2$)
- $\gamma_{\text{small}} \gg \gamma_{\text{large}} \to 0$: small banks face steep wholesale costs
- Regulatory constraint: $e_i \geq \kappa(\omega_M A_i^M + \omega_B A_i^B)$

**Capital buffer:**

$$D_i = e_i - \kappa(\omega_M A_i^M + \omega_B A_i^B) \geq 0$$

Banks with high $\gamma_i \Rightarrow$ large endogenous $D_i$ (self-insurance).

FOCs Back

# First-Order Conditions and the Sectoral Channel

**FOC for asset class $j \in \{M, B\}$:**

$$r_j - r_W - \psi_j A_i^j - \gamma_i (A_i^M + A_i^B - e_i - d_{i,C}) = \lambda_i \kappa \omega_j$$

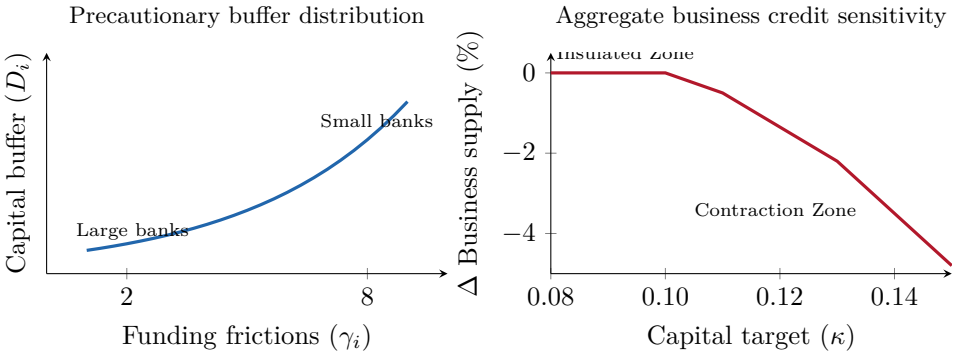
When $\lambda_i > 0$ (constraint binds):

- **Asset Substitution Margin:** $\left|\partial A_i^B / \partial \kappa\right| > \left|\partial A_i^M / \partial \kappa\right|$ because $\omega_B > \omega_M$
- **Scale Deleveraging Multiplier:** $\partial \text{RWA} / \partial A^B = \omega_B = 1 > 0.5 = \omega_M \Rightarrow$ cutting C&I frees 2× the regulatory capital per dollar

**Cross-section:** for low-$\gamma$ banks, $\lambda_i$ stays low $\Rightarrow$ small response. For high-$\gamma$ banks whose buffer is breached, $\lambda_i$ spikes $\Rightarrow$ acute, non-linear contraction in C&I.

Insulated zone Back

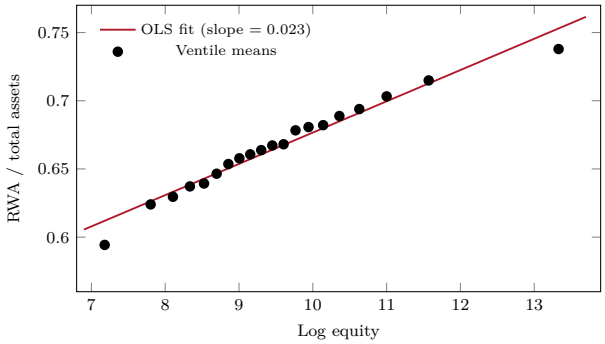
# The Insulated Zone




When shocks are small-to-moderate, well-buffered banks absorb them internally. Only shocks large enough to breach the buffer trigger non-linear credit contractions.

Back

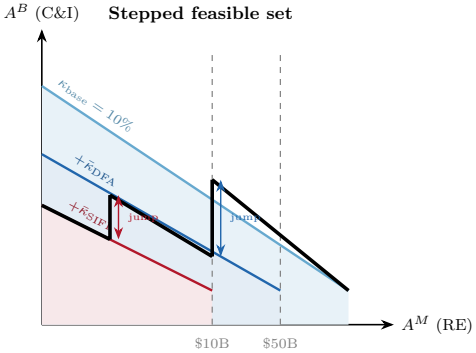
# The Regulatory Wedge



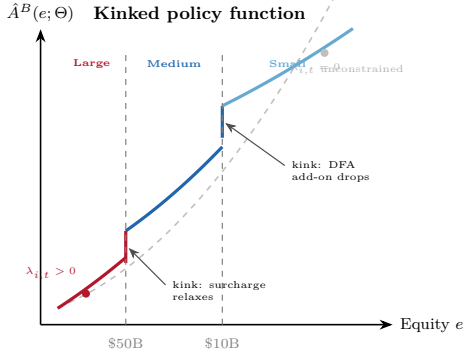

RWA/total-assets ratio rises with bank equity ⇒ larger banks hold higher average risk weights; portfolio composition drives the regulatory position of small banks.

Back

# Why Neural Networks? The Non-Convexity Challenge




$\omega_M A^M + \omega_B A^B \leq e/\kappa_i(\text{Scale})$
Non-convex: frontier jumps at size thresholds




Policy $\hat{A}^B(e;\Theta)$ has kinks at each
tier boundary $\Rightarrow$ perturbation methods fail

Size-tiered surcharges create a stepped feasible set (left) and kinks in optimal policy (right). Standard perturbation around the steady state cannot capture these discontinuities—motivating a global neural-network solution.

# Calibration

| Parameter | Description | Value | Source |
|---|---|---|---|
| $\beta$ | Discount factor | 0.99 | Standard |
| $\delta_e$ | Equity depreciation | 0.001 | Stationarity |
| $\omega_M$, $\omega_B$ | Risk weights | 0.50, 1.00 | Basel III |
| $\kappa_{\text{base}}$ | PCA minimum | 10% | Regulatory |
| $\bar{\kappa}_{\text{DFA}}$ | DFA add-on | 1.5% | Enhanced prudential |
| $\bar{\kappa}_{\text{SIFI}}$ | Large-bank add-on | 2.5% | Stylised |
| $\gamma_{\text{small}}$ | Wholesale funding cost | 0.077 | Funding structure |
| $\gamma_{\text{large}}$ | Wholesale funding cost | $\to 0$ | Funding structure |
| $\psi_M$ | RE adjustment cost | 0.007 | RE share target |
| $\psi_B$ | C&I adjustment cost | 0.05 | Preset |
| $c$ | Cost of capital, *not* risk-weighted | 0.148 | Sectoral aggregates |
| $z$ | Cost of capital, per unit RWA | 0.021 | Jointly with $c$ |
| $\eta_M$, $\eta_B$ | Demand slopes | external | Lit. semi-elasticities |
| $\sigma_{\varepsilon,M}$ | Idiosync. shock (RE) | 0.001 | Charge-off data |
| $\sigma_{\varepsilon,B}$ | Idiosync. shock (C&I) | 0.004 | Charge-off data |

Demand sensitivity
Back to the problem
Back to SMM

# Gross Cost: Framework

**Consumption-equivalent variation (CEV):**

$$V^{\kappa}(\mu^*;\, \kappa^* + 0.01) = V^*((1-\Lambda)\,\mu^*;\, \kappa^*)$$

$\Lambda > 0$ = fraction of steady-state surplus households would forgo to avoid the tightening.

**Decomposition:**

$$\Lambda = \underbrace{\Lambda^{\text{alloc}}}_{\text{sectoral distortion}} + \underbrace{\Lambda^{\text{scale}}}_{\text{credit contraction}}$$

- Two metrics: credit-market surplus and franchise value (discounted dividends)
- **Caveat:** gross cost—no stability benefits, partial equilibrium

Surplus figure

Back

# Backup: Robustness to Sample Extension

Re-estimating small-cohort LPs on the full shock series (2002Q1–2023Q2):

| Treatment | Mean C&I response | Mean RE response |
|---|---|---|
| Baseline (2002–2019) | −0.145% | −0.105% |
| Pandemic dummy × shock | −0.142% | −0.106% |
| Exclude pandemic shocks | −0.142% | −0.106% |

Target moments are essentially unchanged.

Sensitivity to $\psi_B$ Back

# Backup: Sensitivity to $\psi_B$

Constrained-regime sectoral asymmetry $s_B - s_M$ across the admissible range of $\psi_B \in (0.014, 0.142)$:

| $\psi_B$ | $s_B - s_M$ |
|---|---|
| 0.02 | $-3.5$ |
| 0.05 (baseline) | $-2.2$ |
| 0.10 | $-0.8$ |
| Flat weights (any $\psi_B$) | 0.0 |

The asymmetry is a risk-weight phenomenon; $\psi_B$ governs only its amplitude.

Back

# Backup: Precision of the Buffer Gradient

Terciles are estimated as separate local projections, so the *difference* needs a joint covariance: a bank contributes scores to several tercile blocks over its life, and the twenty horizons use overlapping windows.

| | Thin | Thick | Ratio (90% CI) |
|---|---|---|---|
| C&I | −0.1718 | −0.1307 | 1.32 [1.01, 1.62] |
| Real estate | −0.1354 | −0.0964 | 1.41 [1.14, 1.67] |
| Pooled | — | — | 1.37 [1.13, 1.61] |

**The cross-sector restriction is not rejected.** The risk-weight mechanism implies the same gradient in both sectors: measured difference +0.090 (s.e. 0.186), $t = 0.48$, across 9,801 banks.

Back

# Backup: What the Solved Policy Does and Does Not Pin Down

The model is solved by training a policy network. What that choice affects is not what an earlier version of this slide claimed.

**Stable across solved policies.** The welfare cost. With the cost of regulatory capital active, $\Lambda$ lies between 1.16% and 1.42% across solved lending from 0.21× to 1.05× the calibration target — it tracks the *proportional* contraction ($\approx -3\%$), not the level.

**Not stable.** The impulse-response magnitudes, and the buffer gradient when it is recomputed *from* a solved policy rather than from the static problem.

**So the gradient we report is the exact one.** The static problem is quadratic with a linear constraint, so it solves in closed form in each KKT regime: C&I 1.32 against 1.32 in the data. The trained policy returns 1.02–1.06 on the same calibration — it smooths the kink at the frontier, where its residuals are largest — and neither added capacity nor a four-fold longer schedule closes the gap.

What collapses $\Lambda$ to 0.1% is not a low solved level but *removing* $\phi_{i,j}$: without it total credit moves $-0.25\%$ rather than $-3.1\%$. That is a different model, not a numerical sensitivity — and it is the sharpest evidence that $\phi_{i,j}$, not the capital constraint, generates the welfare cost.

Back

# Backup: Market Clearing

**Inverse demand** in each sector:

$$r_{j,t} = \theta_j - \eta_j L_t^j, \qquad j \in \{M, B\}$$

**Clearing** against the solved policies:

$$\tfrac{1}{N} \textstyle\sum_{i=1}^{N} \hat{A}_{i,t}^{j}(\mathbf{S}_{i,t}; \Theta) \;=\; L_t^j(r_{j,t})$$

**Substituting gives a closed-form rate** — imposed analytically at every forward pass, so there is *no numerical inner loop*:

$$r_{j,t}^{*} = \theta_j - \eta_j \cdot \tfrac{1}{N} \textstyle\sum_{i=1}^{N} \hat{A}_{i,t}^{j}(\mathbf{S}_{i,t}; \Theta)$$

Demand calibration Back

# Backup: Market Clearing

**The two schedules are pinned by sample averages and a semi-elasticity:**

$\eta_j = 1/(S_j \bar{L}^j)$ with $S_j$ the demand semi-elasticity; $\theta_j = \bar{r}_j + \eta_j \bar{L}^j$, so the schedule passes through the sample average.

| | mortgages ($M$) | C&I ($B$) |
|---|---|---|
| $\eta_j$ | 6.497e-8 | 3.773e-7 |
| $\theta_j$ | 0.0462 | 0.0847 |
| $\bar{r}_j$ | 0.0154 | 0.0169 |
| $\bar{L}^j$ (USDmm) | 473,578 | 179,618 |
| implied $S_j$ | 32.5 | 14.8 |

Sensitivity Clearing Back

# Backup: Sensitivity to the Demand Calibration

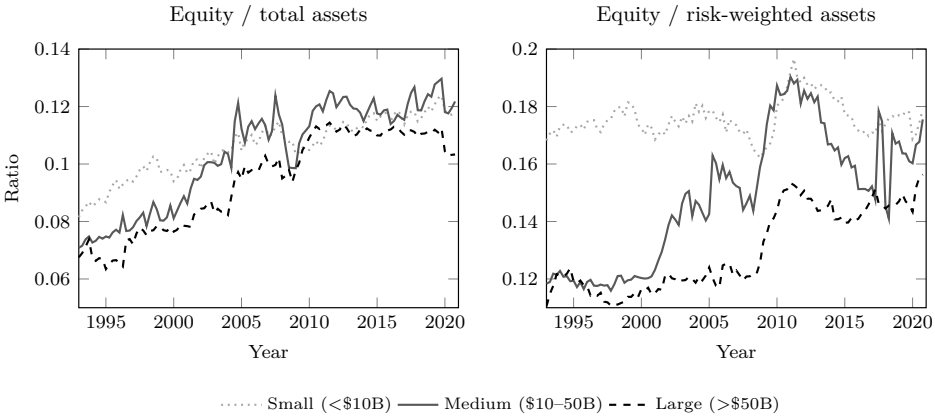
$\eta_j$ enters neither the bank's problem nor the reduced-form estimates, so the lending responses are *exactly* invariant to it. The welfare *level* is not, because the surplus integral is taken under the demand curve.

| $S_M$ | $S_B$ | $\Lambda$ |
|---|---|---|
| 32.5 (implemented) | 14.8 | 1.42% |
| 16.3 | 7.4 | 0.95% |
| 2.0 (cited targets) | 4.0 | 0.27% |

The externally cited estimates measure objects adjacent to a loan-demand schedule — a house-price elasticity, and a loan-*granting* probability against the policy rate — rather than the schedule itself.

Back

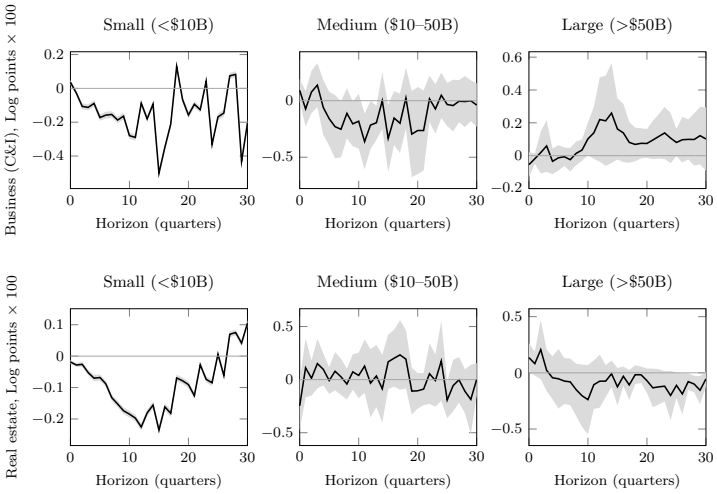
# Backup: Leverage vs. Regulatory Capital Ratios




Small banks appear more capitalised on an unweighted basis, but the advantage compresses under risk-weighting.

Back

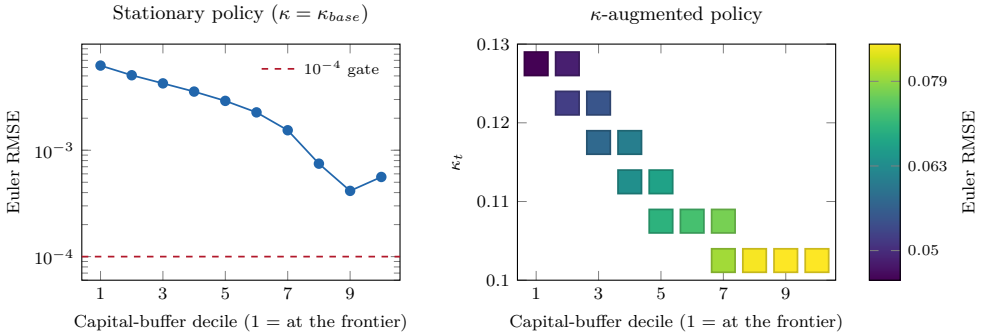
# Backup: Size Cohort Impulse Responses




Small-bank panel precisely estimated; large-bank cohort (≈40 institutions) is not. The buffer distribution, not size per se, drives the cross-section.

Back

# Backup: Solution Accuracy

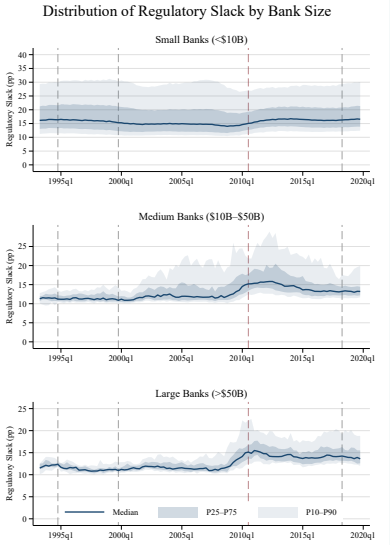



All deciles below $10^{-2}$. Frontier spike reflects the kink in policy functions at the occasionally binding constraint.

Train vs. test | Ergodic coverage

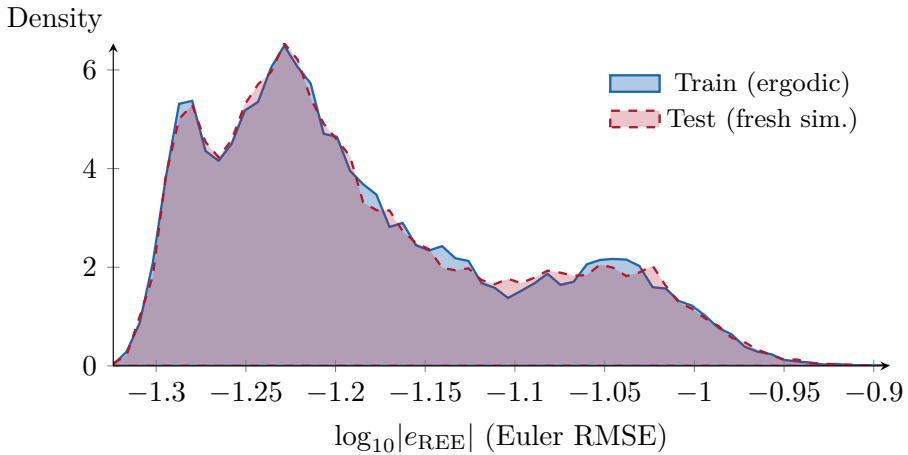
Back

# Backup: Slack Distribution by Size, 1993Q2–2019Q4



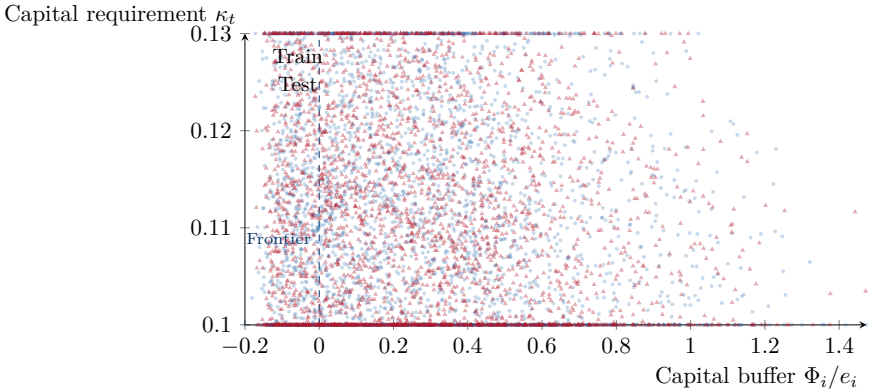

Shaded bands: P10–P90 (light) and P25–P75 (dark). Solid line: median.
Dashed vertical lines mark Riegle-Neal (1994Q4), GLBA (1999Q4), Dodd-Frank (2010Q3, highlighted), and EGRRCPA (2018Q2).
Sample: quarterly bank panel, 1993Q2–2019Q4. Source: US Call Reports.

# Backup: Euler Errors, Training vs. Test




Distribution of $\log_{10}|e_{\mathrm{REE}}|$ on the training panel (seed 0) and an independent test panel (seed 42), 20,000 banks each after 200 periods of burn-in. Median residual $-1.21$ in *both* samples and no fattening of the test tail: the network generalises to unseen states rather than memorising the training trajectory.

# Backup: Ergodic-Set Coverage




Joint distribution of the capital buffer $\Phi_i/e_i$ and the requirement $\kappa_t$, training against test simulation. The test cloud lies entirely inside the training cloud's convex hull, so the policy is evaluated only where it was solved — extrapolation beyond the training support is the most common failure mode of neural-network approximations.

Back

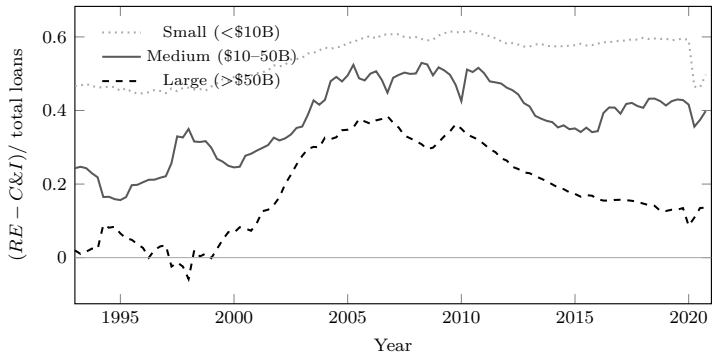
# Backup: Magnitudes in Context

| Study | Design / setting | Effect of +1pp required capital |
|---|---|---|
| Aiyar, Calomiris, and Wieladek (2016) | UK bank-level requirement changes, 1998–2007 | Lending growth 6.5–7.2pp lower |
| Mésonnier and Monk (2015) | EBA 2011–12 capital exercise, euro area | Annualised loan growth 1.2–1.6pp lower |
| De Jonghe, Dewachter, and Ongena (2020) | Pillar-2 decisions, Belgian credit register, 2013–15 | Corporate credit growth $\approx$ 0.2–0.4pp per quarter lower |
| Conti, Nobili, and Signoretti (2023) | Narrative BVAR, Italy (EBA/CA episodes) | Loan supply $-1.5\%$ after 2 years per +40bp Tier 1 ($\approx -3.8\%$ per pp, linearised) |
| This paper (model) | SMM-estimated policy, permanent +1pp $\kappa$ | C&I $-3.5\%$, RE $-3.0\%$, total credit $-3.1\%$ (stationary comparison) |
| This paper (LPs) | Within-bank LPs scaled through the SMM mapping (one shock unit $\mapsto \hat{\sigma}_\kappa = 4.0$bp of $\kappa$) | Trough responses $\approx -8.8\%$ (C&I, $h$=16) and $\approx -5.7\%$ (RE, $h$=12); out-of-sample linear extrapolation |

Model's $-3.5\%$ C&I / $-3.0\%$ RE / $-3.1\%$ total per +1pp sits squarely inside the quasi-experimental range. These magnitudes are conditional on the solved policy; the cross-sectional gradient is not.

Back

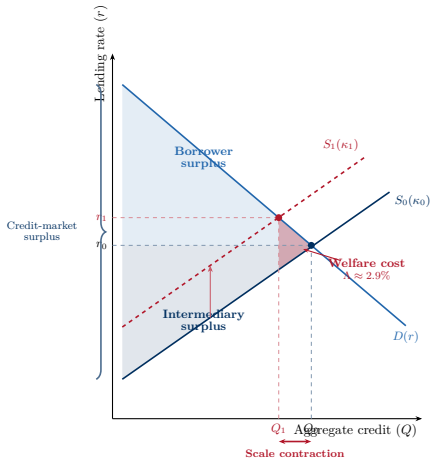
# Backup: Asset Gap




Difference between RE and C&I lending as a share of total loans, by size cohort, 1993–2020.

Back

# Welfare: Credit-Market Surplus

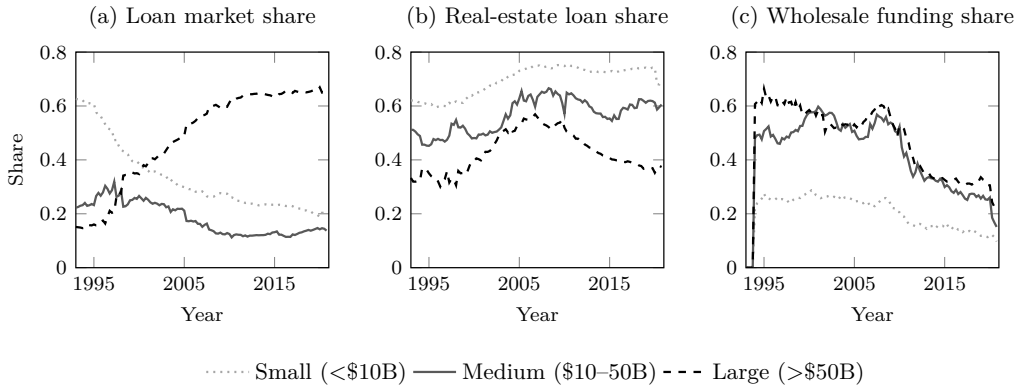



Virtually all of the welfare cost flows through the contraction in the **scale** of intermediation ($\Lambda^{\text{scale}} \gg \Lambda^{\text{alloc}}$)

Borrower + intermediary surplus in the loan market. The red triangle is the welfare cost of a +1pp $\kappa$ tightening; virtually all flows through **scale** contraction ($\Lambda^{\text{scale}} \gg \Lambda^{\text{alloc}}$).

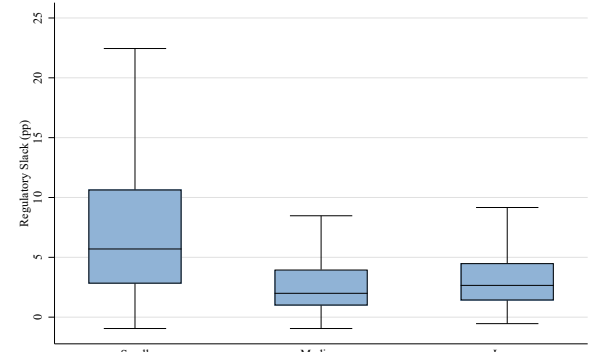
Back

# Backup: Market Concentration and Asset Specialisation




Asset gap Back

# Backup: Buffer Heterogeneity Cuts Across Size Tiers




Source: US Call Reports, 1993Q2–2019Q4.
Regulatory slack: equity-to-RWA ratio minus effective requirement ($\kappa_{it}$), winsorized at 1/99.

Buffers vary as much *within* each statutory tier as across them.

Full distribution | Size-cohort IRFs | Back

# Backup: How Incidence Is Measured

**Burden** = dollars of lending lost. **Incidence** = a group's share of those dollars, against its share of lending:

$$\text{concentration}_g \;=\; \frac{\sum_{i\in g}\Delta A_i \big/ \sum_i \Delta A_i}{\sum_{i\in g} A_i \big/ \sum_i A_i}, \qquad A_i \equiv A_i^M + A_i^B$$

| Buffer tercile | lending | burden | **conc.** | contraction | binding |
|---|---|---|---|---|---|
| Thin | 32.1% | 34.1% | **1.06** | −0.127% | 3.4% |
| Middle | 32.9% | 33.0% | 1.00 | −0.120% | 0.0% |
| Thick | 35.1% | 33.0% | **0.94** | −0.113% | 0.0% |

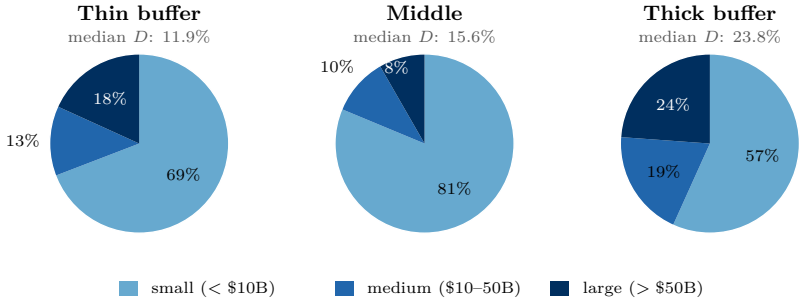
Thin against thick: $1.06/0.94 = 1.13$ — the "13% more per dollar lent" of the previous slide. Thin-buffer banks bear only 6% more than their *own* lending share.

▸ **Scale-free:** every term is a share, so $\sigma_\kappa$, $\rho_\kappa$ and the shock size cancel — incidence is a property of the policy function across the cross-section, not of the shock process

4,000 bank-quarters, +4bp tightening, static problem (closed form in each KKT regime), representative scale.

Back

# Backup: Who Sits in Each Buffer Tercile?

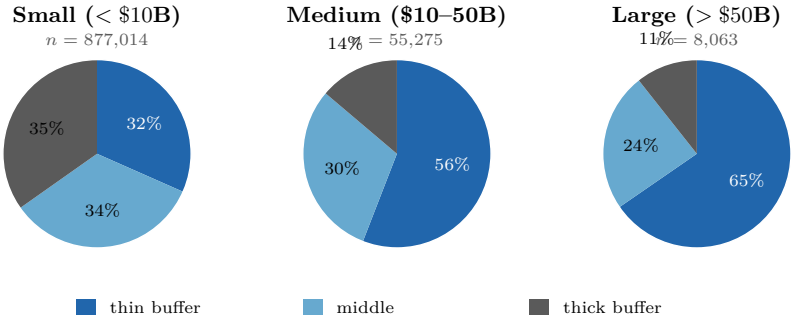



Share of each tercile's *assets* held by banks in each statutory tier. Call Reports through 2019Q4, 20,000 bank-quarters.

- The size does not sort buffers monotonically: the statutory tier is not a proxy for distance to the requirement

Back

# Backup: Buffer Position Within Each Size Tier

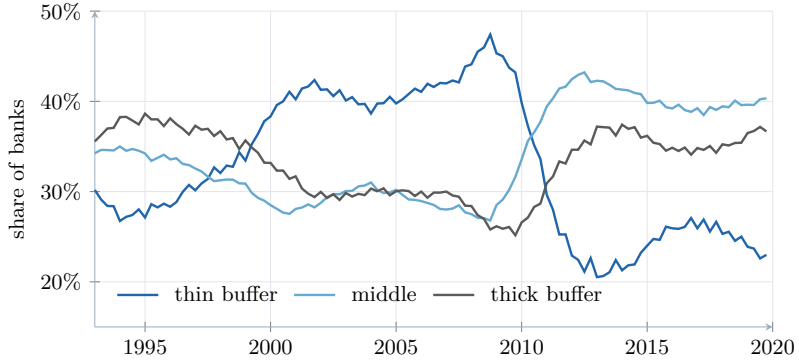



Share of each *size tier's* bank-quarters falling in each buffer tercile. Cutoffs fixed at the pooled 13.65% / 18.47% capital ratio.

- Read down, not across: small banks split almost evenly (32/34/35), but 65% **of large bank-quarters are thin-buffered** and only 11% thick

Back

# Backup: Buffer Composition Over Time




Tercile cutoffs held *fixed* at the pooled 13.65% / 18.47%, so the shares move; 108 quarters, 1993Q1–2019Q4.

- Tercile membership is a *state*, not a bank type: the same cutoffs classify a very different population before and after 2010

Back

# The Regulatory Framework

**Size cohorts by total assets** (Dodd-Frank/EGRRCPA[1] thresholds):

- Community, $<$ \$10B — baseline $\kappa$
- Regional, \$10–50B — $\kappa + 1.5$pp (Dodd-Frank)
- Large, $>$ \$50B — $\kappa + 2.5$pp (enhanced standards)

Concentration Regulatory wedge Regulatory tiers Back

[1] Economic Growth, Regulatory Relief, and Consumer Protection Act (2018), which raised the automatic enhanced-prudential-standards threshold from \$50B to \$250B. The \$50B tier governs almost the whole 1994–2019 sample.

# Backup: Regulatory Tiers, Dodd-Frank and 2019

**Over the estimation sample (Dodd-Frank, 2010–2018)**

| Tier | Trigger | Main requirements |
|---|---|---|
| Community | $< \$10$B | Baseline risk-based minima; PCA well-capitalised |
| Regional | \$10–50B | Company-run stress tests; risk committee |
| Large | $> \$50$B | Enhanced prudential standards: CCAR, supervisory stress tests, liquidity standards, resolution plans |

**After EGRRCPA (2018) and the tailoring rule (October 2019)**

| Category | Trigger | Main requirements |
|---|---|---|
| I | The eight U.S. G-SIBs | G-SIB surcharge, eSLR, TLAC, AOCI in capital, full LCR/NSFR |
| II | $\geq \$700$B assets *or* $\geq \$75$B cross-jurisdictional | Full LCR/NSFR; AOCI in capital |
| III | $\geq \$250$B assets *or* $\geq \$75$B weighted STWF, nonbank assets, or off-balance-sheet exposure | Reduced LCR/NSFR (85%) unless STWF $\geq \$75$B; AOCI opt-out |
| IV | \$100–250B, none of the above | Biennial supervisory stress test; no eSLR; AOCI opt-out |

- Note: category depends on **five** indicators, not size alone — one of them is **weighted short-term wholesale funding**, the margin $\gamma_i$ represents.

G-SIB surcharge Back

# Backup: The G-SIB Surcharge (Category I)

**The surcharge is the *higher* of two scores**, assigned in 50bp buckets:

- **Method 1** (Basel): size, interconnectedness, *substitutability*, complexity, cross-jurisdictional activity — 20% each
- **Method 2** (U.S. only): replaces substitutability with **short-term wholesale funding**, measured as a ratio to **risk-weighted assets**, with *fixed* coefficients

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**Two consequences for this paper:**

- The two margins of the model—wholesale funding and risk weights—enter *one* regulatory indicator, multiplied together
- Substituting C&I ($\omega_B = 1$) into real estate ($\omega_M = 0.5$) lowers RWA, so it *raises* the funding indicator and the surcharge — the opposite sign to the mechanism

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**March 2026 proposal:** index the method-2 coefficients to growth and inflation (they are nominal, so surcharges have drifted up); drop the RWA denominator; reweight STWF to 20% from ≈30%; 10bp buckets. Estimated effect: −3.8% G-SIB CET1.

Back to tiers Back to Data

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